Solution Requirements

Date	25 June 2026
Team ID	xxxxxx
Project Name	Smart Lender – Loan Approval Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviours, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviours (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S. No	**Requirement Category**	**Requirement Description**	**Priority (High/Medium/Low)**
1	Authentication	The system does not require user login. Users can directly access the application and submit loan details for prediction.	Low
2	Authorization Level	The application provides the same functionality to all users. No separate user roles or admin access are implemented.	Low

3	External interfaces	The Flask application loads the trained Machine Learning model (model.pkl) and scaler (scaler.pkl) to generate loan approval predictions.	High
4	Transactions processing	The system accepts applicant details, validates the input, scales the data, processes it through the trained Machine Learning model, and displays the prediction result..	High
5	Reporting	The application displays the loan prediction result on the result page after processing the applicant details.	High
6	Business rules, etc.	Loan approval is predicted based on applicant details such as Gender, Married Status, Dependents, Education, Self-Employed Status, Applicant Income, etc.	High
7	Compliance to laws or regulations.	The application processes user input securely and follows basic data privacy practices.No personal info is permanently stored.	Medium
8	Other	The application is designed to support future enhancements such as additional Machine Learning models, improved prediction accuracy.	Low

Step 2: Non-Functional Requirements (NFR)

S. NO	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate predictions quickly.	Response time less than 3 seconds.
2	Scalability	The system should support increasing users and loan records.	Supports future expansion.
3	Security & Data Privacy	Applicant information should be processed securely during loan prediction. The application does not permanently store user data.	Secure processing of user input.
4	Reliability & Availability	The system should provide consistent prediction results with minimal downtime.	Available 99% of the time.
5	Usability & Accessibility	The interface should be simple and easy for users to navigate.	Easy to use with clear navigation.
6	Others	The system should be easy to maintain and update.	New Features can be added without major changes.